

Unit Test

```
import unittest
```

```
from employee import (  
    PhuongThucTinhThuong,  
    TienthuongNgoaigio,  
    TienthuongNgoaitinh,  
    Laptrinhvien,  
    NhanvienKiemthu,  
    ChuyenvienPhantich,  
    Ketoanvien  
)
```

```
class KiemThuTinhThuong(unittest.TestCase):
```

```
    def test_luong_khac_voi_thuong_co_ban(self):
```

```
        nhan_vien = Laptrinhvien(  
            "LT10",  
            "Hoang Minh Duc",  
            15_000_000  
        )
```

```
        self.assertEqual(  
            nhan_vien.get_tien_thuong(),  
            300_000  
        )
```

```
    def test_thuong_ngoai_gio_voi_luong_cao(self):
```

```
        nhan_vien = NhanvienKiemthu(  
            "TEST20",  
            "Vo Thi Mai",  
            20_000_000  
        )
```

```
nhan_vien.phuong_thuc_tinh_thuong = TienthuongNgoaigio()

thuong = nhan_vien.get_tien_thuong()

self.assertEqual(thuong, 2_000_000)

def test_thuong_ngoai_tinh_voi_luong_khac(self):
    nhan_vien = ChuyenvienPhantich(
        "PT30",
        "Dang Quoc Huy",
        8_000_000
    )

    nhan_vien.phuong_thuc_tinh_thuong = TienthuongNgoaitinh()

    self.assertEqual(
        nhan_vien.get_tien_thuong(),
        1_200_000
    )

def test_thay_doi_phuong_thuc_tinh_thuong(self):
    nhan_vien = Ketoanvien(
        "KT40",
        "Nguyen Thi Hoa",
        10_000_000
    )

    thuong_ban_dau = nhan_vien.get_tien_thuong()

    nhan_vien.phuong_thuc_tinh_thuong = TienthuongNgoaigio()

    thuong_sau_khi_doi = nhan_vien.get_tien_thuong()

    self.assertNotEqual(
        thuong_ban_dau,
```

```

        thuong_sau_khi_doi
    )

    self.assertEqual(
        thuong_sau_khi_doi,
        1_000_000
    )

if __name__ == "__main__":
    unittest.main()

```

```

class KiemThuTinhThuong(unittest.TestCase):
    def test_luong_khac_voi_thuong_co_ban(self):
        nhan_vien = Laptrinhvien(
            "LT10",
            "Hoang Minh Duc",
            15_000_000
        )
        self.assertEqual(
            nhan_vien.get_tien_thuong(),
            300_000
        )
    def test_thuong_ngoai_gio_voi_luong_cao(self):
        nhan_vien = NhanvienKiemthu(
            "TEST20",
            "Vo Thi Mai",
            20_000_000
        )

```

```

PS C:\Users\Lenovo\Downloads> c:\; cd 'C:\Users\Lenovo\Downloads'; & 'C:\Users\Lenovo\AppData\Local\Python\pythoncore-3.14-64\python.exe' 'C:\Users\Lenovo\.vscode\extensions\ms-python.debugpy-2026.6.0-win32-x64\bundle\libs\debugpy\launcher' '54456' '-' 'C:\Users\Lenovo\Downloads\test.py'
PS C:\Users\Lenovo\Downloads> c:\; cd 'C:\Users\Lenovo\Downloads'; & 'C:\Users\Lenovo\AppData\Local\Python\pythoncore-3.14-64\python.exe' 'C:\Users\Lenovo\.vscode\extensions\ms-python.debugpy-2026.6.0-win32-x64\bundle\libs\debugpy\launcher' '50133' '-' 'C:\Users\Lenovo\Downloads\test.py'
PS C:\Users\Lenovo\Downloads> c:\; cd 'C:\Users\Lenovo\Downloads'; & 'C:\Users\Lenovo\AppData\Local\Python\pythoncore-3.14-64\python.exe' 'C:\Users\Lenovo\.vscode\extensions\ms-python.debugpy-2026.6.0-win32-x64\bundle\libs\debugpy\launcher' '65004' '-' 'C:\Users\Lenovo\Downloads\test.py'
....
Ran 4 tests in 0.000s
OK
PS C:\Users\Lenovo\Downloads>

```